

Student Name:



2010

YEAR 12 ASSESSMENT TASK 2

Mathematics Extension 1

General Instructions

- Working Time – 1 1/2 hours
- Write using a blue or black pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a new sheet of paper.

Total Marks - 60

Attempt Questions 1-4

All questions are of equal value.

Question 1 (15 Marks)	Marks
(a) Divide the interval from A (-1, 5) to B (6, -4) <i>externally</i> in the ratio 3:2	2
(b) Simplify $\frac{4^n}{4^{n+1} - 4^n}$	2
(c) Solve the inequality $\frac{2x-3}{x-2} \geq 1$ and graph your solution on the number line	3
(d) Find $\int x\sqrt{1-x} \, dx$ using the substitution $u = 1-x$	3
(e) Use the substitution $t = u^2 - 1$ to evaluate $\int_0^1 \frac{t}{\sqrt{1+t}} \, dt$	3
(f) The curves $y = x^3$ and $y = \frac{x^2 + 3}{4}$ meet at the point (1,1) Find the angle between the tangents to the curves at this point	2

End of Question 1

Question 3 (15 Marks) Begin a new page	Marks
(a) Find the value of k if $\int_0^k \sqrt{1+x} \, dx = \frac{14}{3}$	3
(b) The function $f(x) = x^3 - 5x^2 - 24x + 118$ Taking $x = 5$ as the first approximation of the root of the equation	
(i) Use one application of Newton's Method to obtain a second approximation to a root of the equation.	2
(ii) Explain this result by drawing a sketch of $y = f(x)$ near $x = 5$ stating the sign of $P(5), P'(5), P''(5)$	3
(c) Find the simultaneous solution of $ x-3 < 4$ and $ x-1 > 1$	3
(d) Let $P(x) = -2x^3 + px^2 - qx + 5$	
(i) Show that if $P(x)$ is to have any stationary points then $p^2 - 6q \geq 0$	2
(ii) Explain the situation for $p^2 - 6q = 0$	2

End of Question 3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS.

QUESTION 1

a) $(-1, 5)(6, -4)$

3:2

$$x = \frac{2+18}{1} \quad y = \frac{-10-12}{1}$$

$$= 20 \quad y = -22.$$

$$P(20, -22)$$

$$(b) \frac{4^n}{4^n(4-1)} = \frac{1}{3}$$

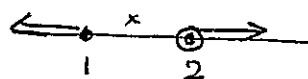
$$(c) \frac{2x-3}{x-2} \geq 1$$

critical point $x=2$.

$$\frac{2x-3}{x-2} = 1$$

$$2x-3 = x-2$$

$$x = 1$$



test $x=0$

$$\frac{-3}{-2} \geq 1 \text{ true}$$

$$x \leq 1, x > 2$$

d) $u = 1-x$
 $\frac{du}{dx} = -1$
 $du = -dx$

$$\int (1-u) \cdot \sqrt{u} \cdot -du$$

$$-\int u^{1/2} - u^{3/2} du$$

$$-\left[\frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right] + C$$

$$-\frac{2}{3} (1-x)^{3/2} + \frac{2}{5} (1-x)^{5/2} + C$$

e) $t = u^2 - 1$
 $\frac{dt}{du} = 2u$
 $dt = 2u \cdot du$
 $t=1 \quad u=\sqrt{2}$
 $t=0 \quad u=1$

$$\int_1^{\sqrt{2}} \frac{u^2-1}{\sqrt{u^2}} \cdot 2u du$$

$$\int_1^{\sqrt{2}} (u^2-1) \times 2 du$$

$$2 \int_1^{\sqrt{2}} (u^2-1) du$$

$$= 2 \left[\frac{u^3}{3} - u \right]_1^{\sqrt{2}}$$

$$2 \left[\left(\frac{2\sqrt{2}}{3} - \sqrt{2} \right) - \left(\frac{1}{3} - 1 \right) \right]$$

$$2 \left[\frac{2}{3} - \frac{\sqrt{2}}{3} \right]$$

$$\frac{4-2\sqrt{2}}{3}$$

f) $y = x^3$ $y = \frac{x^2}{4} + \frac{3}{4}$
 $\frac{dy}{dx} = 3x^2$ $\frac{dy}{dx} = \frac{2x}{4}$
 at $x=1$ $at \ x=1$
 $m_1 = 3$ $m_2 = \frac{1}{2}$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{3 - \frac{1}{2}}{1 + 3 \times \frac{1}{2}} \right|$$

$$= 1$$

$$\therefore \theta = 45^\circ$$

QUESTION 2.

$$\begin{aligned} f(2) &= (2)^3 - 4(2)^2 + 7(2) - 6 \\ &= 8 - 16 + 14 - 6 \\ &= 0 \end{aligned}$$

$\therefore x-2$ is a factor

$$\text{ii) } \frac{x^3 - 4x^2 + 7x - 6}{x-2} = x^2 - 2x + 3$$

this is a quadratic with $\Delta = 4 - 12 = -8$

$\Delta < 0$ no real roots

$\therefore x=2$ is the only real root.

$$\text{b) } f(x) = x^3 + 5x^2 + 17x - 10$$

$$f(0) = -10 < 0$$

$$\begin{aligned} f(2) &= 8 + 20 + 34 - 10 \\ &= 52 > 0 \end{aligned}$$

curve is continuous and there is a sign change so root lies between $x=0$ and $x=2$

ii) use $x=1$

$$\begin{aligned} P(1) &= 1 + 5 + 17 - 10 \\ &= 13 > 0 \end{aligned}$$

root lies between $x=0$ and $x=1$

$$\begin{aligned} \text{iii) } P\left(\frac{1}{2}\right) &= \frac{1}{8} + \frac{10}{8} + \frac{68}{8} - 10 \\ &= \frac{79}{8} - 10 < 0 \end{aligned}$$

The root lies between $x=\frac{1}{2}$ and $x=1$
closer to 1 than to 0

c)

$$\begin{aligned} \text{i) } P(3) &= 27 - 9(k+1) + 3k + 12 \\ &= 27 - 9k - 9 + 3k + 12 \\ &= 30 - 6k. \end{aligned}$$

$$\begin{aligned} \text{ii) If it's divisible then } P(3) &= 0 \\ 30 - 6k &= 0 \\ k &= 5. \end{aligned}$$

$$\begin{aligned} \text{iii) } P(x) &= x^3 - 6x^2 + 5x + 12. \\ &= (x-3)(x^2 - 3x - 4) \\ &= (x-3)(x-4)(x+1) \\ \text{roots } x &= 3, x=4, x=-1 \end{aligned}$$

$$\text{d) } 5^x - 24 = 5^2 - x$$

$$5^x - 24 = \frac{5^2}{5^x}$$

$$5^{2x} - 24 \cdot 5^x - 25 = 0$$

$$\text{let } m = 5^x$$

$$m^2 - 24m - 25 = 0$$

$$(m-25)(m+1) = 0$$

$$m = 25 \quad m = -1$$

$$5^x = 25 \quad 5^x = -1$$

$$x = 2 \quad \text{no solution.}$$

QUESTION 3.

$$a) \left[\frac{(1+x)^{3/2}}{3/2} \right]_0^k = \frac{14}{3}$$

$$\left[\frac{2}{3} (1+x)^{3/2} \right]_0^k = \frac{14}{3}$$

$$\left[(1+x)^{3/2} \right]_0^k = 7$$

$$(1+k)^{3/2} - 1 = 7$$

$$(1+k)^{3/2} = 8$$

$$1+k = 4$$

$$k = 3$$

$$(b) f(x) = x^3 - 5x^2 - 24x + 118 \quad f(5) = -2$$

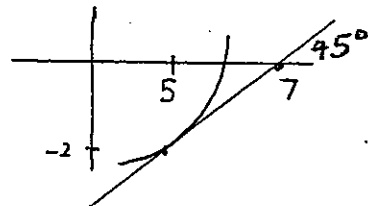
$$f'(x) = 3x^2 - 10x - 24 \quad f'(5) = 1$$

$$x_2 = x - \frac{f(x)}{f'(x)}$$

$$= 5 - \frac{-2}{1}$$

$$= 7$$

$$ii) f(5) \quad f'(5) \quad f''(5)$$



the tangent at $x=5$ cuts the x axis further from the root $x=7$ than first approx $x=5$, due to curvature & concavity. $\therefore x=5$ is better.

$$c) |x-3| < 4$$

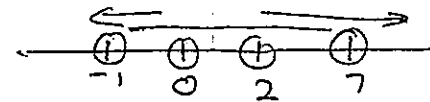
$$|x-1| > 1$$

$$-4 < x-3 < 4$$

$$x-1 > 1 \text{ or } x-1 < -1$$

$$-1 < x < 7$$

$$x > 2 \text{ or } x < 0$$



$$2 < x < 7 \text{ or } -1 < x < 0$$

$$d) i) P(x) = -2x^3 + px^2 - qx + 5$$

$$P'(x) = -6x^2 + 2px - q$$

for stat pt $P'(x) = 0$. with this quadratic

$\Delta \geq 0$ to have real solutions

$\therefore$ for a point to exist $4p^2 - 24q \geq 0$

ii) If $\Delta = 0$
 $x = \frac{-2p}{-12}$
 double root
 so horizon
 inflexion pt

Question 4

a) $1 \times 2 + 3 \times 4 + \dots (2n-1) \times 2n = \frac{1}{3} n(n+1)(4n-1)$

step 1 prove true for $n=1$

$$\begin{aligned} \text{LHS} &= 1 \times 2 = 2 \\ \text{RHS} &= \frac{1}{3} \cdot 1 \cdot 2 \cdot 3 = 2 \end{aligned} \quad \therefore \text{true for } n=1$$

assume true for $n=k$

$$1 \times 2 + 3 \times 4 + \dots (2k-1) \times 2k = \frac{1}{3} k(k+1)(4k-1)$$

prove true for $n=k+1$

$$1 \times 2 + 3 \times 4 + 5 \times 6 + \dots (2k-1) \times 2k + (2k+1) \times 2(k+1)$$

$$= \frac{1}{3} k(k+1)(4k-1) + (2k+1) \times 2(k+1)$$

$$= \frac{1}{3} k(k+1)(4k-1) + 6(2k+1)(k+1)$$

$$= \frac{1}{3} (k+1) [k(4k-1) + 6(2k+1)]$$

$$= \frac{1}{3} (k+1) [4k^2 + k + 6]$$

$$= \frac{1}{3} (k+1) [k+2] [4k+3]$$

$$= \frac{1}{3} (k+1)(k+2)(4(k+1)-1)$$

since it is true for $n=1$ it is true for $n=1+1=2$. since it is true for $n=k$ it is true for $n=k+1$ and hence all positive integer values for n .

b) i) $PS = \sqrt{(2t-0)^2 + (t^2-1)^2}$
 $= \sqrt{t^4 + 2t^2 + 1}$
 $= t^2 + 1$

$PM = t^2 + 1$ (definition of parabola)

$\therefore \Delta PSM$ is isosceles.

ii) $y = \frac{x^2}{4}$

$$\frac{dy}{dx} = \frac{2x}{4}$$

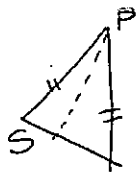
at $P(x=2t$

$$m = t$$

the two lines are perpendicular

$$\begin{aligned} S(0,1) \quad M(2t,-1) \\ m_{SM} &= \frac{1-1}{0-2t} \\ &= \frac{2}{-2t} \\ &= -\frac{1}{t} \end{aligned}$$

iii)



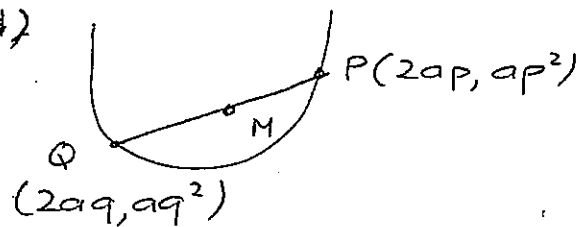
the altitude of isosc Δ bisects vertex

$$\therefore \angle SPC = \angle CPM.$$

also $\angle CPM$ & $\angle APB$ are vert opp & equal

$$\therefore \angle APB = \angle SPC.$$

c)



$$\begin{aligned} \text{i) } M &= \frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \\ &= a(p+q), \frac{a}{2}(p^2+q^2) \end{aligned}$$

ii) equation of PQ

$$y - ap^2 = \frac{ap^2 - aq^2}{2ap - 2aq} (x - 2ap)$$

since through $(0, 4a)$

$$4a - ap^2 = \frac{ap^2 - aq^2}{2a(p-q)} (0 - 2ap)$$

$$4a - ap^2 = \frac{p+q}{2} (-2ap)$$

$$\begin{aligned} 8a - 2ap^2 &= -2ap^2 - 2apq \\ 8a &= -2apq \end{aligned}$$

$$\text{iii) for M } x = a(p+q) \quad y = \frac{a}{2}(p^2+q^2)$$

$$\text{so } \frac{x}{a} = (p+q) \quad y = \frac{a}{2} [(p+q)^2 - 2pq]$$

$$y = \frac{a}{2} \left[\left(\frac{x}{a} \right)^2 - 2x - 4 \right]$$

$$y = \frac{a}{2} \left[\frac{x^2}{a^2} + 8 \right] *$$

$$\underline{\text{OR}} \quad 2y = \frac{x^2}{a} + 8a *$$

$$2ay = x^2 + 8a^2 *$$